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(a)(i)	By Conservation of linear momentum, $0 = m_{\text{Rifle}} v_{\text{recoil}} + m_{\text{bullet}} v_{\text{bullet}}$
	$v_{\text{recoil}} = -2.3 \text{ m s}^{-1}$ (2 sf)
(a)(ii)	Change in momentum of 1 bullet = 0.015 (700 – 0)
	By N2L, Average force by rifle on bullets = $\frac{m v_f - v_i}{\Delta t}$
	= 52.5 N
	By N3L, Average force by bullets on rifle is equal and opposite to average force by rifle on bullets = -52.5 N
	Alternative Change in momentum of rifle = 10(4.5)(-2.33333 – 0)
	By N2L, Average force by bullets on rifle = $\frac{m v_f - v_i}{\Delta t}$
	= -52.5 N
(b)(i)	Initial KE = 10 J Final KE = 5 J
	Since initial KE is greater than final KE, collision is inelastic.
	Alternative Relative speed of approach = 15 m s^{-1} Relative speed of separation = 10 m s^{-1}
	Since relative speed of approach is not equal to relative speed of separation, the collision is inelastic.
(b)(ii)	Impulse = change in momentum
	= -2.0 N s
(b)(iii)	If KE is fully converted to other forms of energy at the same time during the collision, total / sum of momentum is zero.
	By conservation of momentum, since total / sum of initial momentum is not zero, this is not possible.
(c)(i)	From point of release of block B to the point of contact with spring: Net loss of GPE of blocks is converted to gain in KE of blocks.
	From point of contact with spring to point where net force on block B is zero: Net loss of GPE of blocks is converted to gain in KE of blocks and EPE of spring.
	From point where net force on block B is zero to point of maximum compression: Net loss of GPE and KE of blocks is converted to gain in EPE of spring.
(c)(ii)	Net loss in GPE = Gain in EPE
	$k = 460 \text{ N m}^{-1}$ (2 sf)

